

Q. $\Rightarrow$ Solve the following system of linear eq using crammer's Rule:

$$\begin{aligned} x + 2y + 3z &= -5 \\ 3x + y - 3z &= 4 \\ -3x + 4y + 7z &= -7 \end{aligned}$$

$$\begin{matrix} \downarrow x \\ \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & -3 \\ -3 & 4 & 7 \end{bmatrix} \end{matrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -5 \\ 4 \\ -7 \end{bmatrix}$$

$\rightarrow$ Step 1, Find first Determinant:

$$|D| = 40$$

Now using crammer's rule for finding the value of x , y and z :

$$x = \frac{|D_x|}{|D|}, \quad y = \frac{|D_y|}{|D|}, \quad z = \frac{|D_z|}{|D|}$$

So find D_x :

$$|D_x| = \begin{vmatrix} -5 & 2 & 3 \\ 4 & 1 & -3 \\ -7 & 4 & 7 \end{vmatrix} =$$

$|D_x|$: Determinate of $|D_x| = -80$.

$$\frac{|D_x|}{|D|} = \frac{-80}{40} = -2$$

$x = -2$
same for others